

HCI's emerging roles in shaping alignment discretion

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Abstract

Due to the wide application of foundation models across society, aligning these systems with human values, preferences, and safety has become critical. Although it is well established that human desires, values, and preferences inherently conflict, the role of interfaces collecting human feedback in quietly shaping which values and preferences get encoded in AI systems remains understudied. This paper argues that Human-Computer Interaction (HCI) plays an active, underrecognized role in shaping what Buyl et al. [8] call alignment discretion: the interpretive latitude annotators exercise when alignment principles such as helpfulness, safety, and honesty inevitably conflict. We make two initial contributions. First, we propose a bidirectional view of alignment discretion, establishing that algorithmic outputs co-constitute how annotators perceive, deliberate, and decide, not only the other way around. Second, we argue that annotation interfaces actively construct rather than neutrally transmit this discretion, opening critical research directions at the intersection of HCI, bidirectional AI alignment, and data annotation. This exploratory and theoretical paper proposes four emerging roles for HCI in alignment discretion, suggesting that HCI plays a more prominent role in shaping alignment quality and legitimacy than is currently recognized, indirectly shaping the values embedded in AI systems.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

bidirectional alignment, alignment discretion, human-AI alignment

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1 Introduction

Consider an annotator, choosing between two AI responses to a request for persuasive advice: one is helpful but manipulative, the

other safe but generic. With seconds to decide, their choice, along with many thousands like it, is shaping how AI systems such as Large Language Models (LLMs) balance helpfulness against harm, between truthfulness and quick user satisfaction [33].

Aligning AI with human values and preferences has received increasing scholarly attention [12, 22, 34]. Despite growing interest in AI-generated alignment data [22], recent evidence demonstrates the continued necessity of high-quality human expert annotations [17, 26, 41]. Contemporary AI alignment increasingly requires domain expertise rather than generic "data labeling". Industry reports document growing demand for annotators with specialized knowledge in law, medicine, and other critical fields, making nuanced safety and trustworthiness judgments [19]. This shift makes understanding how experts exercise judgment in alignment work not merely academically interesting, but practically urgent.

Expert judgments, preferences, and values do not exist as simple, consistent hierarchies. Alignment work therefore involves complex ethical deliberations and competing principles, tensions that constitute one of the central, understudied challenges of human-AI alignment [14, 16, 44]. This complexity is what makes alignment discretion unavoidable. Recently, Buyl et al. [8] identify a critical yet underexamined phenomenon at the heart of this effort: "alignment discretion", the interpretive latitude annotators must exercise when alignment principles or values conflict. Commonly used principles in AI alignment like "be helpful" and "be safe" inevitably conflict. This invites risks: annotators may exercise discretion arbitrarily rather than systematically, and models may fail to learn it faithfully. As a result, problematic value structures can enter AI systems blindly, deployed in critical domains from medicine to education, rendering alignment both unmeasurable and potentially misaligned with its stated goals. Yet, current dominant alignment approaches, including Reinforcement Learning from Human Feedback (RLHF) [30], provide little support to resolve these conflicts, leaving the exercise of discretion largely unexamined and unsupported.

A recent turn in alignment indicates that human-AI alignment operates bidirectionally rather than unidirectionally: humans shape AI through specification and feedback, while AI influences humans in turn, requiring cognitive and behavioral adaptation for meaningful alignment [34]. We extend this insight to alignment discretion itself. Discretion, we argue, is equally bidirectional, co-constituted through interaction. Annotators produce training data that shapes model preferences, while model behavior, algorithmic outputs, and annotation interfaces may mediate annotation efforts, reshaping how annotators deliberate, and judge. This turn is consequential: it begins to reposition the annotation interface, not merely as peripheral scaffolding, but as something closer to a constitutive presence in AI alignment. Early work already demonstrates that even simple

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differences in interface guidance, for basic and advanced annotators for instance, can affect annotation quality [17], suggesting that interfaces play an underexplored role in how alignment discretion is exercised.

We make two initial theoretical contributions. First, we frame alignment discretion as bidirectional, not something annotators simply exercise, but something that flows in both directions: annotator judgment shapes AI behavior, while model outputs and interface design reshape how annotators deliberate and decide, with implications for how alignment research is understood. Second, we argue that annotation interfaces do not merely transmit this discretion: they actively construct it. What judgments can be expressed, which cognitive biases are amplified, whether careful deliberation is possible, is shaped by design decisions of these interfaces. This realization motivates four exploratory research directions for how HCI can support more effective alignment discretion: HCI for (1) human-AI deliberation guidance, (2) for metacognitive support, (3) for rendering discretion visible, (4) for moral imagination. We acknowledge the limitations of HCI alone in the complex sociotechnical system of human-AI alignment: it cannot dissolve the labor structures or economic incentives that constrain annotation work. But discretion is exercised through interfaces, and understanding how those interfaces shape it is, we argue, essential to understanding alignment and its practices.

2 Related work

This paper sits at the intersection the following research streams: AI alignment, where annotation practices determine what values enter AI systems and how, bidirectional human-AI alignment, which reconceives that relationship as mutually constitutive and how HCI influences this relationship. We review each in turn before introducing the concept, alignment discretion, that connects them in the next section.

2.1 AI alignment and RLHF

Due to the wide application of foundation models across society, aligning these systems with human values, preferences, and safety requirements became critical [12, 22, 35]. Reinforcement Learning from Human Feedback (RLHF) has become a dominant approach to AI alignment [30]. Typically, RLHF encodes human preferences in AI systems through a three-stage process. A base model is first fine-tuned on demonstrations of preferred behavior. Annotators compare pairs of model-generated responses and indicate which they prefer; these judgments train a reward model that learns to predict human preferences. Finally, the language model is optimized through reinforcement learning to maximize that reward. Consider the annotator from our opening, faced with a response that is helpful but manipulative, and one that is safe but generic. In RLHF, their choice is recorded as a single preference signal. What produced it, the reasoning about harm, the weighing of helpfulness against honesty, does not. That deliberation is invisible to the reward model and, therefore, to the system it trains.

However, expert judgment in annotation is exercised under conditions that offer little support for it: annotators consistently prioritize surface features such as fluency over factual accuracy [33], and even basic interface differences measurably affect judgment

quality [17]. The interfaces through which this work is done are treated as neutral conduits. They are not examined as shapers of the discretion being exercised.

2.2 HCI's role in annotation interfaces

The role of HCI in alignment annotation is exercised primarily through data annotation interfaces, which allow annotators to compare and evaluate model outputs for preference collection. These interfaces typically present annotators with multiple AI-generated responses, such as pairwise comparisons of helpfulness, or provide structured evaluation mechanisms, including rating scales, meta-data labels, and ranking tasks [1, 4, 30]. Recent work has sought to improve these annotation practices, for instance, through modular interface systems such as RLHF-Blender [29] and approaches based on interactive decomposition [36], which aim to enhance annotation quality and efficiency. However, these efforts largely focus on optimizing task execution, while paying limited attention to supporting annotators in identifying and reasoning about the underlying values and principles reflected in AI outputs. How these approaches can support effective alignment discretion remains unexplored.

2.3 Bi-directional alignment

A growing body of work challenges the assumption that alignment flows in one direction only. Shen et al. [34] review over 400 papers to argue that alignment must encompass both the conventional direction, ensuring AI produces intended outcomes, and an underexamined inverse: supporting humans in cognitively and behaviorally adapting to AI. Their systematic review identifies significant gaps in long-term interaction design, human value modeling, and mutual adjustment dynamics, concluding that treating alignment as a static, unidirectional process systematically underestimates what the human side of the equation requires. We turn this perspective to alignment practices themselves. What has not been examined is alignment discretion, the judgment annotators exercise when principles conflict, from a bidirectional perspective. This re-framing and its consequences are the questions this paper reflects on.

3 Alignment discretion as bidirectional

3.1 What is alignment discretion?

Buyl et al. [8] identify alignment discretion as the interpretive latitude annotators exercise emerging when principles conflict: helpfulness versus safety, autonomy versus harm prevention, confidence versus nuance. This discretion is unavoidable in RLHF because: multiple legitimate principles exist, no resolution principle determines how conflicts should be resolved, and concrete cases require situated judgment. Therefore, the question is not whether discretion exists, but what forms it takes. Buyl et al. [8] identify two key risks when discretion is exercised without care: “(i) annotators may exercise discretion arbitrarily rather than systematically, and (ii) models may fail to capture the discretion faithfully.” They argue that legitimate discretion requires transparency about decision-making processes and mechanisms for oversight, much like judicial discretion operates within institutional checks. Without deep understanding and clarity on this discretion, it is unclear what AI systems are aligned to. Therefore, we cannot measure if we are

successful, resisting meaningful evaluation critical for alignment research [8].

In practice, this discretion is exercised through hundreds of small choices and clicks, minor preference selections, one-on-one interactions situated between the annotator and the RLHF interface. With every choice, discretion is performed, yet currently these choices often go unexamined and annotators receive little support for meaningful discretion through the interface. Empirical evidence makes this especially troubling: annotators consistently prioritize superficial features like fluency over factual accuracy [20], and LLM explanations that appear credible frequently contain errors that go unchallenged [37]. Yet current research on alignment discretion does not account for this interactive nature, leaving unexplored how discretion is actively constructed through human-AI interactions.

3.2 Towards a bidirectional view of alignment discretion

Human and algorithmic outputs are not static but unfold through ongoing interactions. For instance, AI-driven recommender systems not only reflect but also shape the preferences they ostensibly measure [31], as users' interactions with recommendations iteratively influence both their future preferences and the system's subsequent outputs. In this way, human preferences and values are continuously shaped and reshaped through interaction with algorithmic outputs. We extend this notion of human-AI coevolution [31] into annotation practice itself. In the alignment loop, AI outputs condition what annotators attend to; annotation decisions, in turn, condition what AI systems learn to produce. Each reconfigures the other, quietly, iteratively, continuously.

While identifying value tensions in alignment discretion is critical, a growing body of research shows that humans have limited capacity to detect the values and principles at stake in AI outputs. Empirical evidence demonstrates that users fail to recognize privacy leakage in AI system [45] and allow AI confidence to inappropriately calibrate their own [23]. If humans detect these tensions poorly or are swayed by AI's confident presentation, alignment discretion is prone to be arbitrary or unreliable. Addressing this requires attending not only to how AI learns from humans, but also to how interfaces support human judgment and mitigate these cognitive limitations.

The interactive complexity is further magnified by Glickman and Sharot [15], who show that social biases intensify in human-AI interactions, exceeding those observed in human-only settings. While not situated in alignment contexts, their findings suggest that bidirectional dynamics do not merely transmit human characteristics but can amplify certain patterns, such as biases, while suppressing others. This raises a critical question: how do these interactive dynamics shape the discretion exercised in alignment annotation?

A bidirectional alignment perspective shifts the focus from treating human and algorithmic alignment discretion separately, to examining how these interact and co-evolve. While humans shape AI through value specification and feedback, alignment also requires supporting humans in adapting to AI—cognitively, behaviorally, and critically engaging with model outputs [34]. We therefore argue that effective alignment requires a bidirectional perspective on alignment discretion.

Viewing alignment discretion in this way foregrounds the role of interaction: how annotators exercise discretion, how models learn from it, and how these processes reshape one another over time. This shift in perspective has practical implications: because this bidirectional process occurs through annotation interfaces, these interfaces come to play a more prominent role in alignment practices, positioning HCI interfaces as critical sites where discretion interaction manifests. Thus, understanding alignment discretion from this perspective requires scrutiny from both AI alignment and human-computer interaction, with the latter playing a more critical role than currently recognized.

4 HCI's emerging roles in alignment discretion

Viewing alignment discretion as bidirectional shifts attention to annotation interfaces as a material site where alignment discretion is enacted and constructed. HCI is well positioned to address this: it has a long tradition of studying human-AI interaction, its mutual impacts, and the design interventions that shape it [3, 10, 42]. Yet this body of work has not been turned toward alignment discretion specifically. Prior work already demonstrates the consequences of neglecting HCI perspectives in annotation, over- and under-trust, ineffective evaluation, and poor annotation quality [2, 28, 36, 43]. How HCI can support the quality and legitimacy of discretion, the interpretive judgments at the heart of alignment, remains largely unexplored. We outline four exploratory directions that begin to map this space, while acknowledging that HCI is only part of the sociotechnical machine of human-AI alignment.

4.1 HCI for human-AI deliberation guidance

Effective discretion necessitates deliberation. Unlike judges, who exercise discretion within established legal procedures and deliberative traditions, alignment annotators work through annotation platforms, evaluation dashboards, and feedback interfaces that offer little equivalent support. Annotators, as alignment decision-makers, need to navigate what Dworkin [11] calls "dimensions of weight," weighing principles and values like harmfulness and helpfulness. Traditional RLHF interfaces offer limited support for step-by-step reasoning, partial disagreement, or thoughtful deliberation.

Drawing on recent HCI literature, we identify two complementary approaches to supporting human-AI deliberation. The first focuses on assisting humans in their own deliberation process through step-by-step interfaces and inviting written explanations for their annotations. For instance, decomposition principles, which break down judgments into individual claims, show promising results [36]. Expanding this to discretion, taking inspiration from legal deliberation processes, could support more effective alignment discretion. HCI interfaces might guide annotators in identifying which principles are at stake, articulating their tensions, documenting their reasoning, and expressing degrees of confidence and uncertainty. Second, we identify approaches that support interactive deliberation with AI assistants. Prior work suggests that structured forms of discussion between humans and AI can facilitate communication around conflicting values. For instance, Ma et al. [27] demonstrate that interactive conversations aimed at identifying differences in viewpoints between humans and AI reduce users' tendency to over-rely on AI suggestions through iterative decision updates and

structured exchanges. Such approaches enable more flexible, conversational interactions and more precise information exchange with domain-specific models. More broadly, their initial results suggest that step-by-step interfaces, qualitative documentation, guided deliberation, and the explicit highlighting of viewpoint differences can support more effective alignment decision-making. However, how these approaches interact, and whether they scale to real-world annotation contexts, remain open questions.

4.2 HCI for metacognitive support

Humans exhibit cognitive biases in AI-assisted decision-making [6], tend to over- and under-trust AI recommendations, and show amplified perceptual and cognitive biases in human-AI interactions compared to human-to-human interactions [15]. Moreover, Sidra et al. [38] argue that metacognitive thinking in human-AI collaboration is increasingly critical as traditional metacognitive heuristics fail to transfer from human-only contexts. We therefore argue that effective alignment deliberation requires metacognitive support for annotators, support that accounts not only for inherent human cognitive limitations but for the additional biases that human-AI interaction itself introduces.

We identify two initial strategies through which HCI can provide metacognitive support. First, visual guidance that reduces cognitive load by directing attention and explicitly highlighting differences. Earlier work [17] shows that interfaces providing visual feedback, color-coded annotation overviews, and time locks preventing hasty submissions helped annotators engage more carefully with complex cases, reducing cognitive burden. Similarly, Mocha [13] uses visual highlighting to make counterfactual differences explicit. DeBiasMe [24] visualizes potential biases in both prompts and AI outputs to heighten user awareness on potential biases. These examples suggest potential for visual support that counteracts cognitive biases and makes differences between options explicit.

A second strategy is to pursue deliberate friction that prompts critical reflection. Benedetti and Mauri [5] propose design interventions that deliberately introduce pauses for reflection, stimulate independent thinking over automation, and help users appropriately to calibrate trust [7]. DeBiasMe [24], for instance, experiments with interface friction to reduce bias amplification by making users aware of anchoring bias in their prompts, while Clark et al. [9] theoretically propose anti-sycophancy measures for human-AI alignment. Other empirical research with AI users demonstrates that temporal friction, such as time delays that slow down interactions in dialogues, can improve performance on multi-domain tasks in goal-oriented conversations [21], suggesting that temporal friction might similarly support more careful deliberation in annotation contexts. Following, we propose to build on existing work on biases in human-AI interactions [6] to identify which cognitive biases are most problematic for effective alignment discretion and which arise specifically from the intersection of human and algorithmic discretion. Thereby, applying these metacognitive support strategies to alignment discretion, through visual guidance and friction that supports critical thinking, introduce promising directions.

4.3 HCI for rendering discretion visible

Making alignment discretion visible is essential for both individual improvement and systemic oversight. Buyl et al.'s research [8] on legitimate alignment discretion emphasizes the necessity of transparency about decision-making processes and mechanisms for oversight. Current RLHF interfaces typically capture only final binary preferences, losing the reasoning behind alignment choices. Documentation interfaces could support three interconnected goals. First, individual learning where annotators review past decisions, identify their own patterns, and calibrate judgment over time. Second, collective learning where teams, both human and AI, discuss how different annotators resolve similar conflicts and build shared understanding of when to prioritize competing values. Third, oversight and accountability through audit trails showing how discretion was exercised and enabling intervention when it becomes arbitrary. Design approaches might include annotator journals for explaining complex decisions, comparative dashboards showing how different annotators approached similar cases, or reflection prompts that periodically summarize patterns and prompt metacognition. In this way, documentation could not merely record discretion but shapes how it can be reflected upon, learned from, and standardized. Expanding these to capture not just what was chosen but including the reasoning behind it, the why, represents a critical direction for supporting effective discretion.

4.4 HCI for moral imagination

Finally, we make explicit a fourth role annotation interfaces may shape the very space of possible moral reasoning available to annotators. Current alignment practices, particularly in RLHF, often reduce complex value conflicts to constrained decision formats such as pairwise comparisons or scalar ratings. While scalable, these formats risk producing what Schwerzmann and Campolo [32] describe as a focus on “desired behaviors” and corrective mechanisms, narrowing the space of moral possibilities during deliberation. This raises a deeper concern: not only how annotators decide, but what they are able to imagine as legitimate alignment outcomes. Schwerzmann and Campolo push beyond behavioral correction in alignment, toward tools that help “make life worth living in our technological milieus” [32], a vision that current annotation practices rarely reach. Helm and Gerlek press further still, calling for AI systems open to “pluriversal figurations of technology” [18]: not just better values, but a wider horizon of whose values can be expressed.

Drawing on the notion of moral imagination, the capacity to consider alternative perspectives, value configurations, and possible consequences in ethical decision-making—annotation interfaces can be understood as shaping the boundaries of this imaginative space. From this perspective, interface design does not merely guide deliberation but actively participates in constructing what forms of discretion become possible. Future work might explore whether annotation interfaces can support moral imagination by expanding discretionary possibilities beyond binary choices that may replicate existing value hierarchies in RLHF [32], and by leveraging design mechanisms as a form of ethical inquiry and a form of prototypical kind of ethical thinking [25, 39]. This could involve enabling annotators to propose alternative responses, articulate competing value framings, or explore counterfactual scenarios, for instance through

approaches such as Overton pluralism [40]. Such interventions may support a more pluralistic and reflective form of alignment [40]. However, how such interfaces can be designed in practice, and how expanded expressive capacity interacts with scalability constraints, remain open questions for future research.

4.5 Four exploratory HCI roles in alignment discretion

The four roles interact and depend on one another. Deliberation requires time and structured processes. Metacognitive support enables more deliberation by surfacing what matters, what is imagined, and reducing cognitive bias. Documentation shapes what can even be deliberated about, by making otherwise invisible patterns available for reflection. Together, they sketch a design space, one that current alignment practice largely leaves unexplored. A critical limitation remains that interaction design alone cannot redistribute power when economic structures require speed. Better interfaces under old incentive structures remain constrained. The point is not that HCI solves alignment discretion, but that it shapes what alignment discretion is deemed possible.

5 Discussion and future research

Viewing alignment discretion as bidirectional reframes both alignment practices and HCI's role in them. Rather than treating human discretion and algorithmic discretion as separate phenomena, bidirectionality reveals them as dynamically co-adaptive: annotator discretion shapes training data, models learn from these data, and model behavior potentially reshapes how annotators understand and exercise discretion. This interaction can create risks beyond those Buyl et al. [8] identify: certain discretion patterns may be amplified at the human-AI interface while others are suppressed, biases may compound through iteration, and which values are perpetuated in AI systems is influenced by these dynamics. If annotation interfaces actively construct discretion rather than transmitting it, then interface design becomes a critical site for alignment work. The choices HCI practitioners make, what can be expressed, how much time is available, what learning support exists, how deliberation is structured, indirectly shape what values get encoded into AI systems. This positions HCI not as a peripheral usability concern but as a core contributor to human-AI alignment quality and legitimacy. This perspective opens research directions. How do algorithmic and human discretion co-constitute one another across annotation cycles? How do the interfaces through which alignment work is performed construct, enable, or constrain this discretion? How can the four exploratory HCI roles concretely support more effective and imaginative alignment discretion?

6 Conclusion

Alignment discretion, the latitude annotators exercise when principles conflict, is critical in alignment research. When not handled carefully, it embeds arbitrary values in widely used AI systems. We argue that arbitrary alignment discretion does not emerge in isolation but is co-constructed bidirectionally through interactions between interfaces, annotator judgment, and reward models. Together, these dynamics open a research direction at the intersection

of AI alignment, HCI, and annotator work, a space this paper begins to map.

This paper has made two initial contributions. First, we extend Buyl et al.'s [8] identification of alignment discretion into a bidirectional frame, examining how discretion is not merely exercised but actively constructed through human-AI interaction. Second, we argue that annotation interfaces are not neutral conduits for human judgment. They can shape what forms of reasoning and judgment become thinkable in the first place. They are, in this sense, sites where AI alignment is co-constituted.

From this bi-directional perspective, we have proposed four exploratory roles through which HCI could support more effective alignment discretion. These do not resolve the structural and economic pressures that constrain annotation work, but they make visible a design space that is currently underexplored. We position HCI not as a peripheral usability concern but as an indirect contributor to alignment quality and legitimacy, one that bears on what values and preferences get encoded into AI systems, and whose judgment counts in that process. We invite empirical work to explore and contest these roles. Alignment discretion is not peripheral to the challenge of building AI systems that genuinely reflect human values, it is constitutive of it. And in shaping how that discretion unfolds, HCI plays a more prominent role than is currently recognized.

7 Limitations

This work is preliminary and exploratory. We theorize bidirectional dynamics but do not demonstrate them empirically. The proposed HCI roles require empirical validation and refinement through practice. We focus on interfaces while acknowledging the broader context of labor conditions, organizational incentives, and power structures that shape annotation work. Interface design is necessary but not sufficient for effective alignment discretion.

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